

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from matplotlib.colors import ListedColormap
```

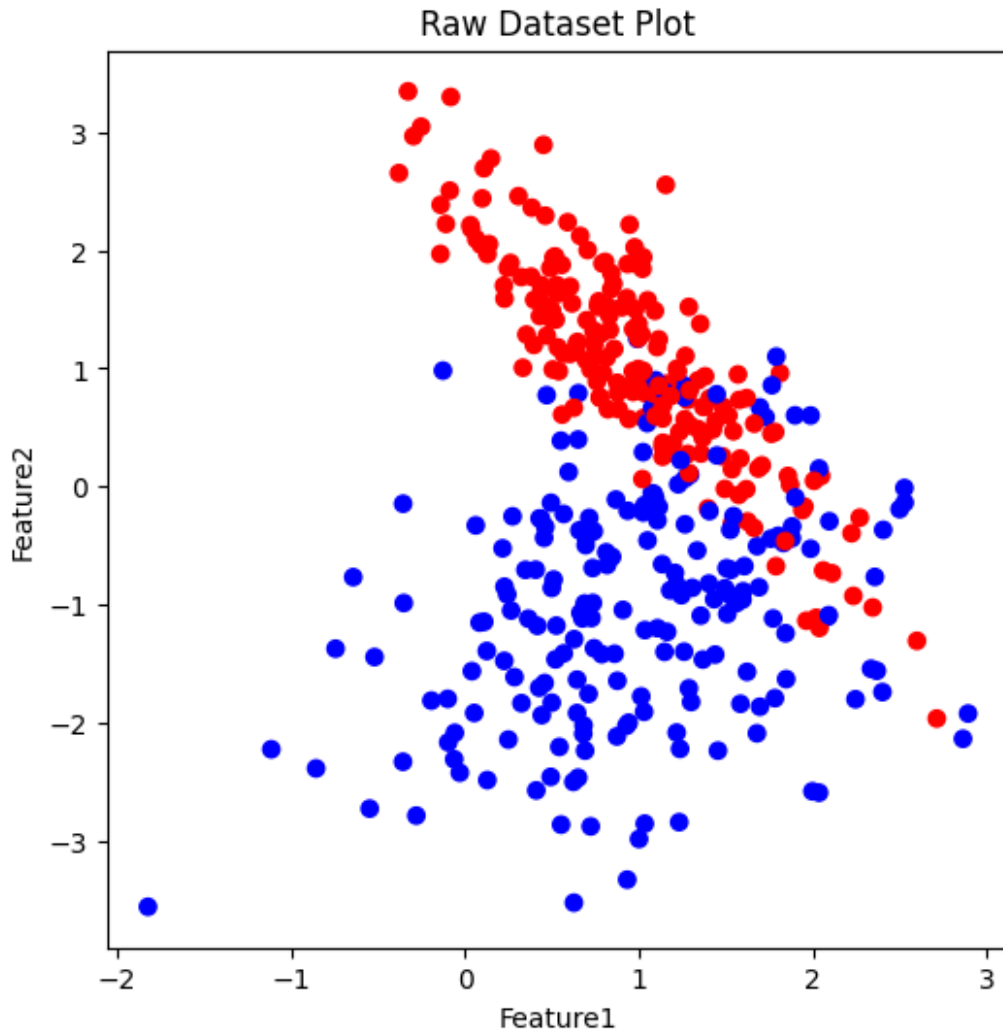
```
df = pd.read_excel("random_forest_dataset.xlsx")
X = df[["Feature1", "Feature2"]].values
y = df["Label"].values
```

```
print(df.head())
print("Dataset Shape =", X.shape)
```

	Feature1	Feature2	Label
0	1.431868	-0.953095	0
1	0.323473	-1.833522	0
2	0.977841	1.501977	1
3	1.986525	0.602109	0
4	1.356117	0.280041	1

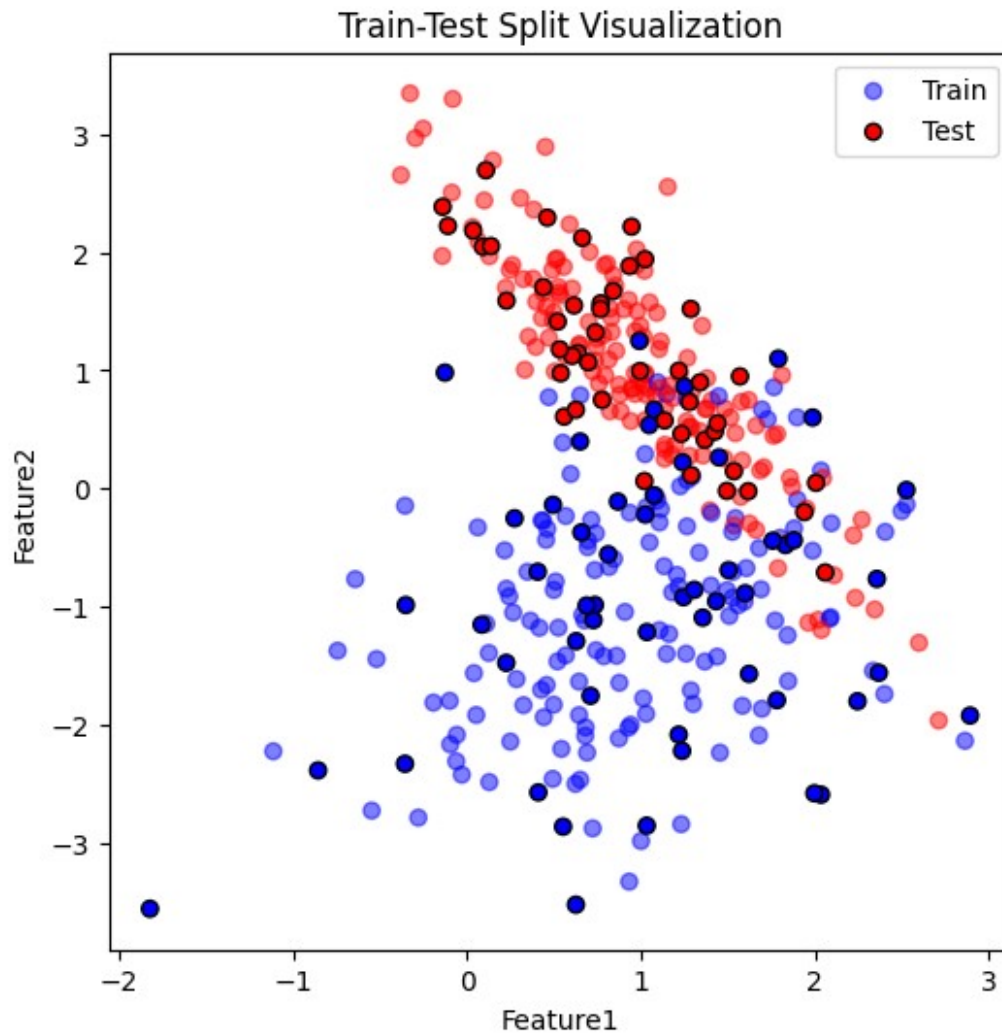
Dataset Shape = (400, 2)

```
plt.figure(figsize=(6,6))
plt.scatter(df["Feature1"], df["Feature2"], c=df["Label"], cmap="bwr")
plt.title("Raw Dataset Plot")
plt.xlabel("Feature1")
plt.ylabel("Feature2")
plt.show()
```



```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=42
)

plt.figure(figsize=(6,6))
plt.scatter(X_train[:,0], X_train[:,1], c=y_train, cmap="bwr",
            alpha=0.5, label="Train")
plt.scatter(X_test[:,0], X_test[:,1], c=y_test, cmap="bwr",
            edgecolor='k', label="Test")
plt.title("Train-Test Split Visualization")
plt.xlabel("Feature1")
plt.ylabel("Feature2")
plt.legend()
plt.show()
```



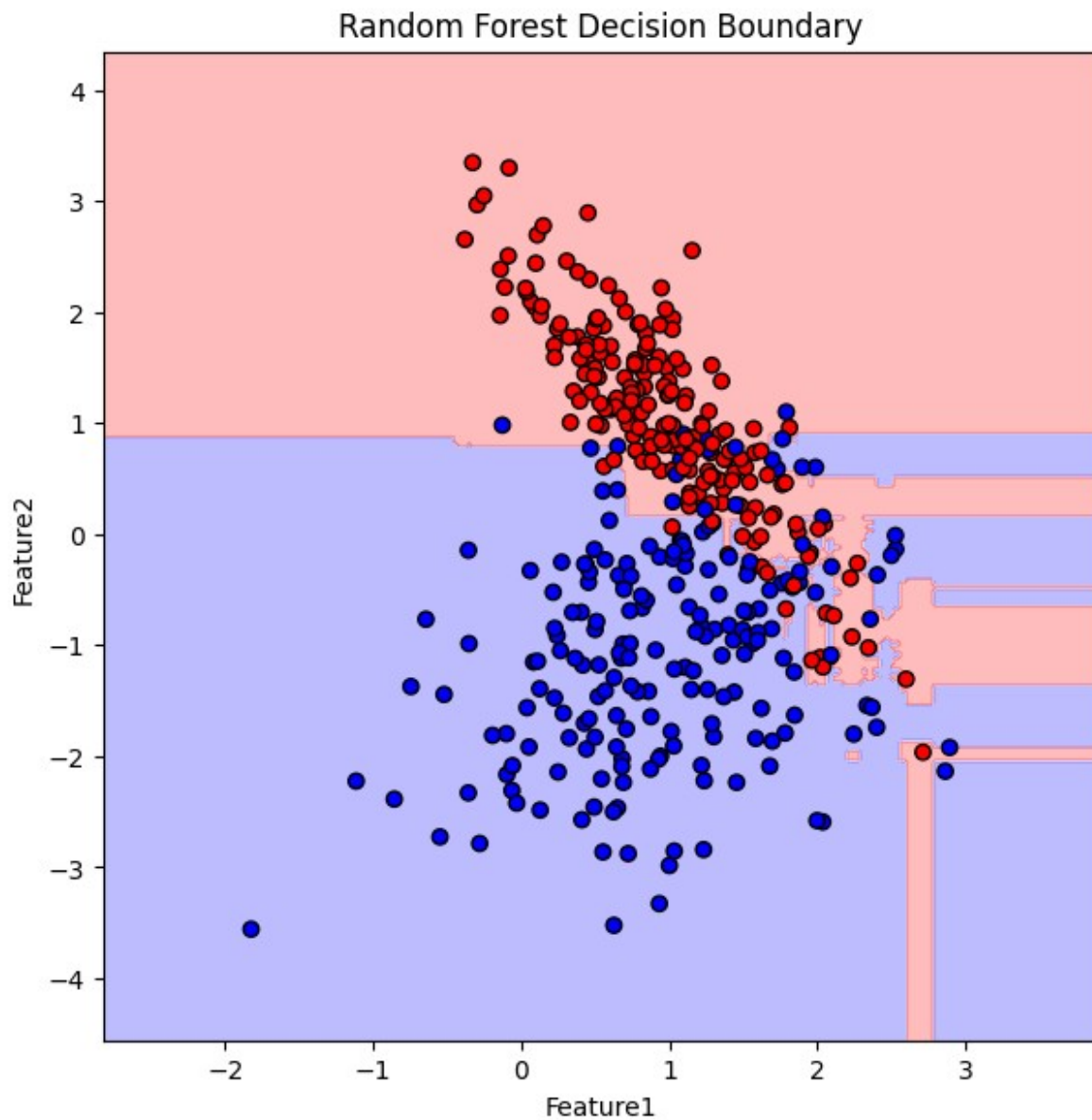
```
rf = RandomForestClassifier(  
    n_estimators=50,  
    criterion="gini",  
    random_state=42  
)  
  
rf.fit(X_train, y_train)  
print("Random Forest Model Trained Successfully!")  
  
Random Forest Model Trained Successfully!  
  
x_min, x_max = X[:,0].min()-1, X[:,0].max()+1  
y_min, y_max = X[:,1].min()-1, X[:,1].max()+1  
  
xx, yy = np.meshgrid(  
    np.linspace(x_min, x_max, 300),  
    np.linspace(y_min, y_max, 300)  
)
```

```

Z = rf.predict(np.c_[xx.ravel(), yy.ravel()])
Z = Z.reshape(xx.shape)

plt.figure(figsize=(7,7))
plt.contourf(xx, yy, Z, alpha=0.3, cmap="bwr")
plt.scatter(X[:,0], X[:,1], c=y, cmap="bwr", edgecolor='k')
plt.title("Random Forest Decision Boundary")
plt.xlabel("Feature1")
plt.ylabel("Feature2")
plt.show()

```



```
importances = rf.feature_importances_  
features = ["Feature1", "Feature2"]  
  
plt.figure(figsize=(6,4))  
plt.bar(features, importances, color=["blue", "red"])  
plt.title("Feature Importance (Random Forest)")  
plt.ylabel("Importance Score")  
plt.show()
```

